



Late Pleistocene to Holocene paleoenvironmental evolution of Lake Hazar, Eastern Anatolia, Turkey



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ABSTRACT

High-resolution seismic reflection profiles and core analyses in Lake Hazar provide a detailed record of the lake level fluctuations and the robust chronology of paleoclimatic events of the Eastern Anatolia during the late Pleistocene to Holocene. The earlier period of Marine Isotope Stage 3 (MIS-3) prior to 48 cal ka BP was accompanied by considerable lake level drop below −95 m, whereas the lake level increase during the later period (ca. 48–29 cal ka BP) gave rise to deposition of a transgressive unit with typical of aggradational architecture in the seismic reflection profiles. High climate variability with the Greenland interstadials and stadials (Dansgaard-Oeschger and Heinrich events) in Lake Hazar are sensitively recorded in the core sediments by using multi-proxy analyses. Adjustment of seismic units correlating with the radiocarbon-dated chronostratigraphic units in the studied cores implies that the early Marine Isotope Stage 2 (MIS-2) is marked by another lowstand lake level existed at ca. −90 m during 29–23 cal ka BP. In comparison to MIS-3 stage, the multi-proxy analyses reveal a general dry evaporative condition during most of the Last Glacial Maximum. In Lake Hazar, the maximum humidity in the late glacial period existed during 14.9–13.5 cal ka BP. The existence of a hiatus in the sedimentary record is documented in the seismic data that coincides with the cold and dry Younger Dryas period, implying a remarkable lake level drop. The multi-proxy records of the Holocene sediments reveal that a maximum precipitation in the early Holocene period prevailed during 10.1–9.3 cal ka BP, leading water level rise in the lake. In Lake Hazar, the middle Holocene until 4.9 cal ka BP is represented by highly climate variations, indicating a series of shorter wet and longer dry climate periods. The late Holocene is accompanied by lake level rises under a general wet climate condition that was interrupted by short dry climate intervals during 3.7–3.3 cal ka BP, 2.8–2.6 cal ka BP and 2.1–1.8 cal ka BP.

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1. Introduction

Lake sediments provide a continuous, sensitive record of changing conditions and processes within the lake itself and in the surrounding catchment, thus they contain significant archives of past environmental change on the continents. Valuable information on regional and global scale patterns of past climate variability in the Mediterranean region have been documented by

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numerous paleoclimate studies (Bar-Matthews et al., 2000; Allen et al., 2002; Martrat et al., 2004; Drysdale et al., 2005; Jones et al., 2007; Tzedakis, 2007; Zanchetta et al., 2007; Kotthoff et al., 2008; Schmiedl et al., 2010), but some of these may partly explain the contradictory palaeoclimate histories. Although paleoclimate changes in the eastern Anatolia during the late Pleistocene to Holocene are well constrained in terms of proxy-climate data (Kempe et al., 2002; Wick et al., 2003; Jones et al., 2006; Litt et al., 2009; Kuzucuoğlu et al., 2011; Eriş, 2013; Çağatay et al., 2014), there are still ambiguities on the timing and intensity of specific climate periods (e.g., Last Glacial Maximum, Heinrich event 1, Bolling-Allerod and Younger Dryas) resulting in difficulties in correlating

different Anatolian lake records. However, the magnitude, nature, and duration of each period indicate geographical variation, therefore, further records are needed to establish a complete picture of the climatic conditions in the eastern Anatolia.

Lake Hazar occupies at the southeast of Elazığ in the eastern Turkey, and is the deepest natural lake along the East Anatolian Fault Zone (EAFZ) (Figs. 1 and 2). Although considerable work has been carried out on the EAFZ (Akkan, 1972; Hempton and Dunne, 1984; Çetin et al., 2003; Aksoy et al., 2007; Garcia-Moreno et al., 2010), very limited exposed terraces in the southern coastal outcrops with absence of radiocarbon ages render precise determination of the lake level changes (Erinç, 1953; İnandık, 1965). The Lake Hazar basin has been described previously as a pull-apart basin (Arpat and Şaroğlu, 1972; Hempton and Dunne, 1984; Şengör et al., 1985; Çetin et al., 2003), or a negative flower structure (Aksoy et al., 2007) on basis of onland field studies. The first detailed morpho-tectonic investigation of Lake Hazar has been carried out by Garcia-Moreno et al. (2010), who presented geophysical and sedimentological data from the lake. In their study, the structure of the basin is analyzed utilising bathymetric maps and seismic profiles, suggesting a pull-apart basin evolution for the Lake Hazar basin. Although the paleoclimate and lake level changes in Lake Hazar have been recently documented by Eriş (2013) for the last 15 ka BP, the lack of age control could not provide a promising chronostratigraphy of the late Pleistocene to Holocene sedimentary record.

To remedy this privation, we revised previous preliminary high resolution seismic reflections and two piston cores to provide a detailed record of sedimentation in Lake Hazar during the late Pleistocene to Holocene. In the present study, we used a robust age-depth model for the last 18 cal ka BP (Ön et al., this issue) with additional AMS ^{14}C ages obtained from the previously studied core by Eriş (2013). We also analyzed a new core to extend the Lake Hazar sedimentary record until ca. 48 cal ka BP. Multi-proxy analyses of the cores serve to investigate the high-resolution of

paleoclimatic and water level changes in Lake Hazar, thus, we are able to compare these climate oscillations with those records obtained from the other Anatolian lakes.

2. Study area

2.1. Regional setting

Lake Hazar with a modern 1255 m altitude is a 7 km wide and 25 km long intra-montane depression in the eastern Turkey (Figs. 1 and 2). It is one of the well-developed tectonic lake basin located on the East Anatolian Fault Zone (EAFZ) that is a left-lateral strike-slip fault trending NE-SW. Lake Hazar comprises a rhomboidal shaped deep basin in the east with a maximum depth of -220 m (Garcia-Moreno et al., 2010; Eriş, 2013) that is bordered by two strike-slip segments of the EAFZ (Fig. 2). The southern coastline of the lake is very straight and fault controlled, and delimited by the Mount Hazar (2350 m) uplift, whereas the northern coastline is unrelated to any faulting. According to onshore observations (Çetin et al., 2003; Garcia-Moreno et al., 2010), Gezin (GF) and Havri (HF) faults around the Lake Hazar basin constitute the sub-parallel left-lateral strike-slip fault strands (Fig. 2). These branches of the EAFZ are named from north to south as the Master Fault (MF) and the Southern Fault (SF) (Garcia-Moreno et al., 2010). The seismic data reveals another fault branch, that was previously defined as the Northern Fault (Garcia-Moreno et al., 2010). The bedrock in the western catchment area of Lake Hazar consists of Elazığ Magmatics of Senonian age containing the plutonic rocks (Yazgan and Chessex, 1991; Robertson et al., 2007). The northeast and southeast of the lake are covered by Hazar Group of Maastrichtien-Upper Paleocene age which consists of the conglomerate limestone, sandy limestone and gray-brown shales. The volcanic and volcano-clastic rocks of the Maden Complex of the middle Eocene age overlying the Hazar Group is well-exposed all around the lake (Erdem, 1987).

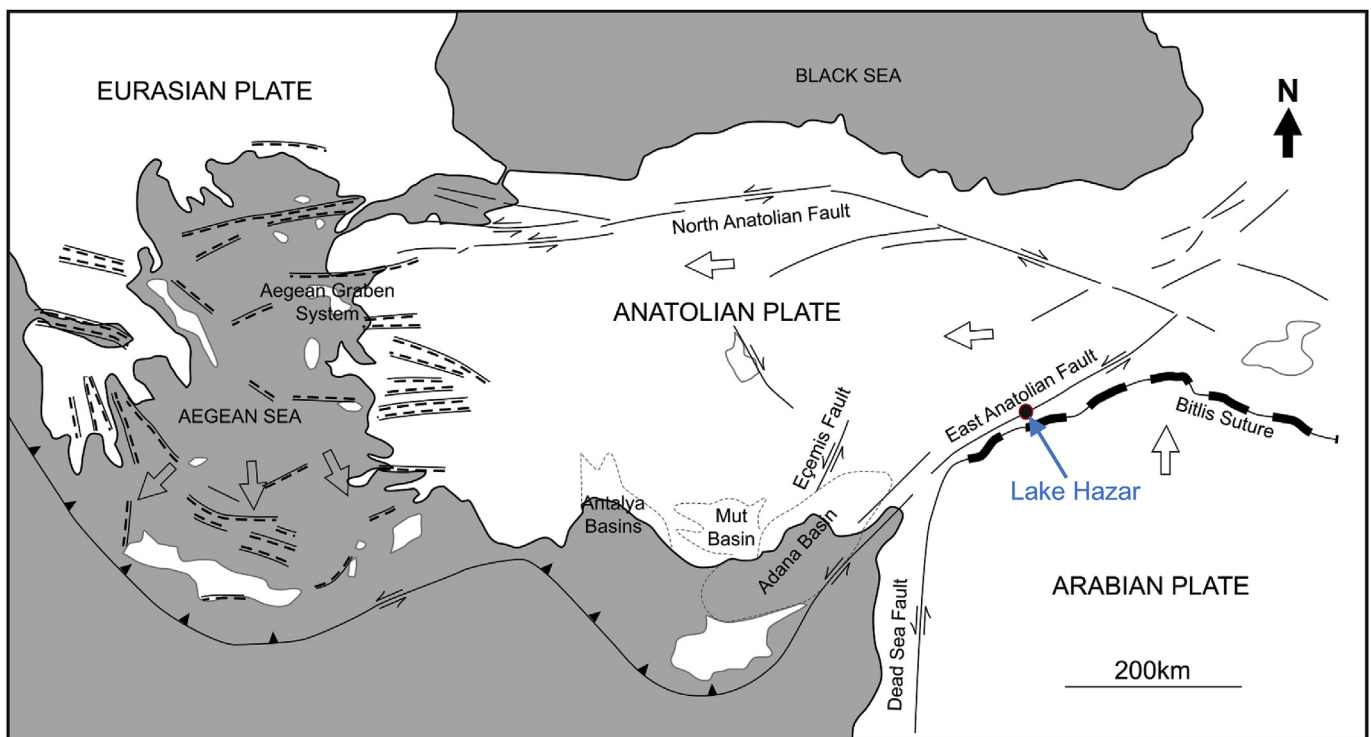


Fig. 1. Simplified map showing location of Lake Hazar in the eastern Turkey.

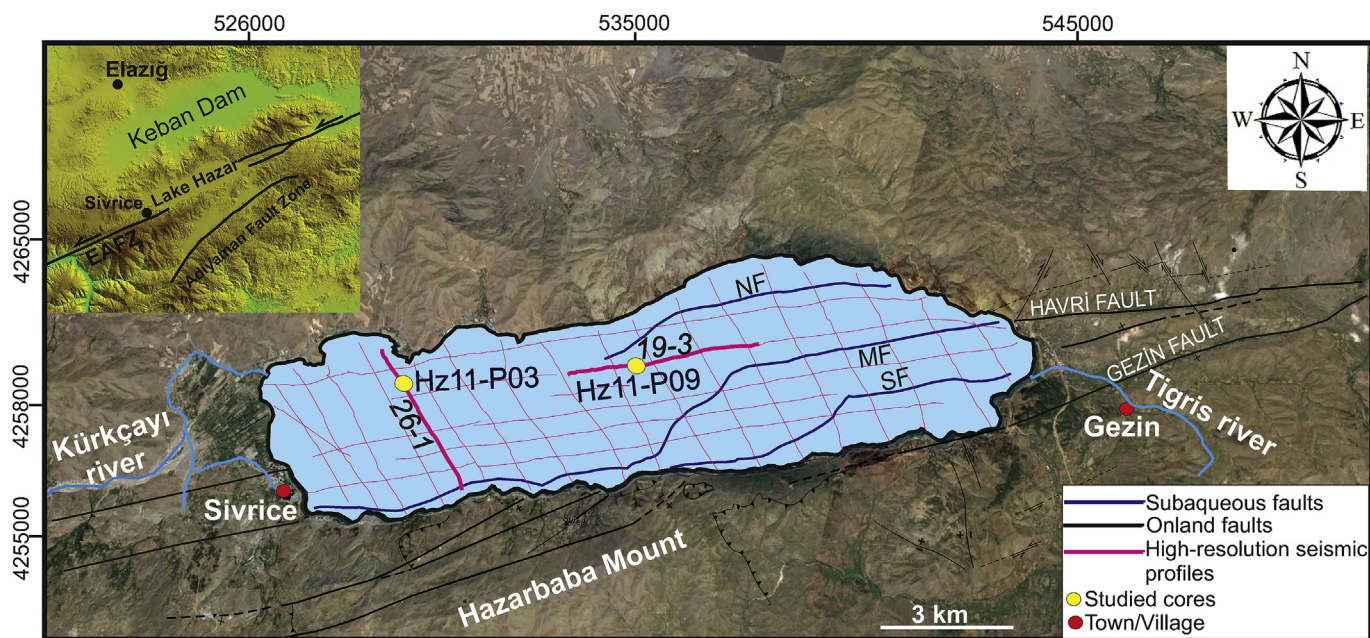


Fig. 2. A satellite map showing the studied high-resolution seismic profiles (red lines) and the sediment cores (yellow dots) retrieved from Lake Hazar. The blue lines crossing the lake are the subaqueous fault segments that formed in relation with the East Anatolian Fault Zone in the region. Kürkçayı River at the western extreme of the lake is biggest present-day drainage around the lake. Hazarbaba Mount (2350 m) is the highest catchment area bordering the southern coast of the lake. MF: Master Fault; SF: Southern Fault; NF: Northern Fault. The black lines are the onland faults according to Çetin et al. (2003). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

2.2. Current climate and hydrography

Several studies were carried out on the limnology and the water chemistry of Lake Hazar (Şen et al., 2002; Ünlü et al., 2008; Kirat and Bölücek, 2010; Saler et al., 2011). The surface area of the lake is 274 km² and the volume is 7.5×10^9 m³ (Fig. 2). Chemical weathering of magmatic rocks, evaporation, and hydrothermal activity create high alkalinity with a pH of 8.75, and a salinity of 16 per mill (Ünlü et al., 2008). Today, Lake Hazar is a superficially closed hydrological system. The southern shore of Lake Hazar is bounded by steep slopes of the surrounding mountains associated with dense drainage networks (Fig. 2). Several permanent streams and ephemeral creeks draining the surrounding catchments provide a high amount of water and sediments to the lake. The biggest drainage system is the Kürkçayı River in the west, which forms the largest present-day Kürkçayı delta (Fig. 2). The Lake Hazar region represents a continental climate with an average temperature of the region is about 0.7 °C in January and about 25.3 °C in July (Kirat and Bölücek, 2010; Saler et al., 2011). The highest precipitation occurs during winter and spring. Precipitation exceeds evapo-transpiration from October to April, whereas the reverse is the case for the remaining part of the year (Şen et al., 2002).

3. Materials and methods

3.1. Seismic reflection and core data

High-resolution seismic reflection profiles in Lake Hazar were collected in 2012 by using a INNOMAR SES 2000 high-resolution sub-bottom profiler system (Eriş, 2013). The data from the tow-fish, the sub-bottom profilers and the navigation system were recorded in XTF format. Depths of reflectors were calculated using average water and sediment velocities of 1450 m s⁻¹ and 1550 m s⁻¹, respectively. We incorporated one published profile (line 26-1) by Eriş (2013), whereas one additional reflection profile

(line 19-3) is examined for the first time in the present study (Figs. 2 and 3). In contrast to Eriş (2013), the original navigation seismic files were imported to the Kingdom Suite software for seismic interpretation. Thus, adjustment of seismic units correlating with the radiocarbon-dated chronostratigraphic units in two studied cores executed by precisely matching core depth with sub-bottom depth in seismic profiles.

The studied two cores were collected by a percussion piston corer (7.5 cm diameter) during two different lake surveys in 2011 and 2012 (Fig. 2; Table 1). The core stations were selected on the basis of stratigraphic informations obtained from the chirp profiles (Fig. 3). Core Hz11-P03 with a 3.48 m length was previously described (Eriş, 2013), whereas a new piston core Hz11-P09 having a 3.84 m length is examined in the present study (Fig. 4). It was transported to the Core Analyses Laboratory of the İstanbul Technical University, Eastern Mediterranean Centre for Oceanography and Limnology (ITU-EMCOL), and split into two halves. The working core halves were sampled at 5 cm resolution for mineralogic and TOC analyses.

3.2. Geochemical and physical analyses

In this study, μ -X-Ray Fluorescence (XRF) element analysis was carried out at 3 mm resolution with an ITRAX core scanner in ITU-EMCOL Geochemistry Laboratory. The relative elemental abundances of Fe, Ca, Sr and Ti were recorded as counts per second (cps). Ca is most likely derived from endogenic calcite that is related to productivity into the water column (Cohen, 2003), whereas high Fe and Ti elements are attributed to high clastic input to lakes (Fischer and Wefer, 1999). Thus, the Ca/Ti ratio served through the studied cores for the purpose of normalization with respect to carbonate or siliciclastic sedimentation. The enrichment of Sr relative to Ca indicates that the composition of lake sediments is dominated by aragonite, rather than calcite. The total organic carbon (TOC) content for samples was analyzed using a Shimadzu TOC/TIC analyser

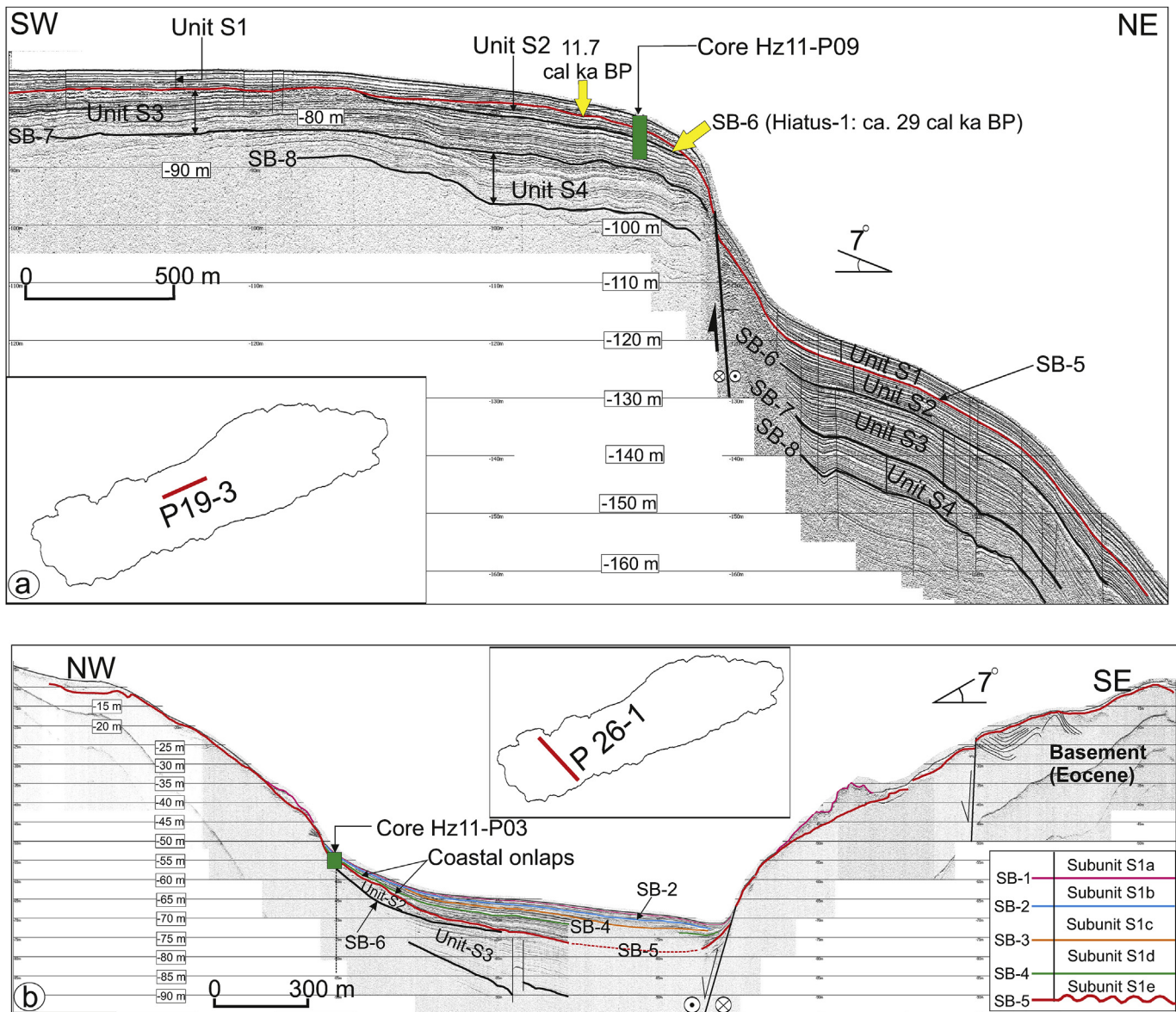


Fig. 3. (a) SW-NE high-resolution seismic profile P19-3 showing the late Pleistocene to Holocene sedimentary units in Lake Hazar from S4 to S1 (for location of profile, see Fig. 2). Core Hz11-P09 penetrates surface SB-6 into the sediments of Unit S3. The reflection surface SB-6 corresponds to a hiatus in core Hz11-P09. The major unconformity surface (red reflection; SB-5) truncating the overall late Pleistocene units (S4, S3 and S2) is dated at 11.7 cal ka BP. (b) NW-SE high-resolution seismic profile P26-1 showing the late Pleistocene to Holocene sedimentary units in Lake Hazar (revised after Eriş, 2013) (for location of profile, see Fig. 2). In the profile, the transgressive Holocene unit S1 is recognized by a typical of onlapping sediments onto the major unconformity surface (red reflection), and was previously subdivided into five subunits (L1e to L1a) in the seismic profile. Core Hz11-P03 penetrates surface SB-5 into the sediments of Unit S2. The unconformity surface SB-5 corresponds to a hiatus in core Hz11-P03 (see Fig. 4), implying a subaerial erosion existed at the core depth (–53 m). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

at geochemistry laboratory. The TOC is calculated as the difference between TC and TIC contents. The TOC contents of the core sediments characterize the amount of organic matter (OM) in the sediment (Burdige, 2006).

Magnetic susceptibility (MS) measurements were performed at

1 cm resolution by a Bartington point sensor of Geotek Multi-Sensor Core Logger equipment in İTÜ-EMCOL Core Analysis laboratory. Magnetic susceptibility (MS) provides a useful proxy indicator of changes in the relative intensity of terrigenous input into the lake. The dolomite content of samples obtained from core Hz11-

Table 1
Core informations for this study.

Core name	Core length (m)	Water Depth(m)	Coordinates (UTM)	Seismic Lines
H211-P03	3.48	–53	529385 E 4258886 N	Profile 26-1
H211-P09	3.84	–81	534254.13 E 4261403.52 N	Profile 19-3

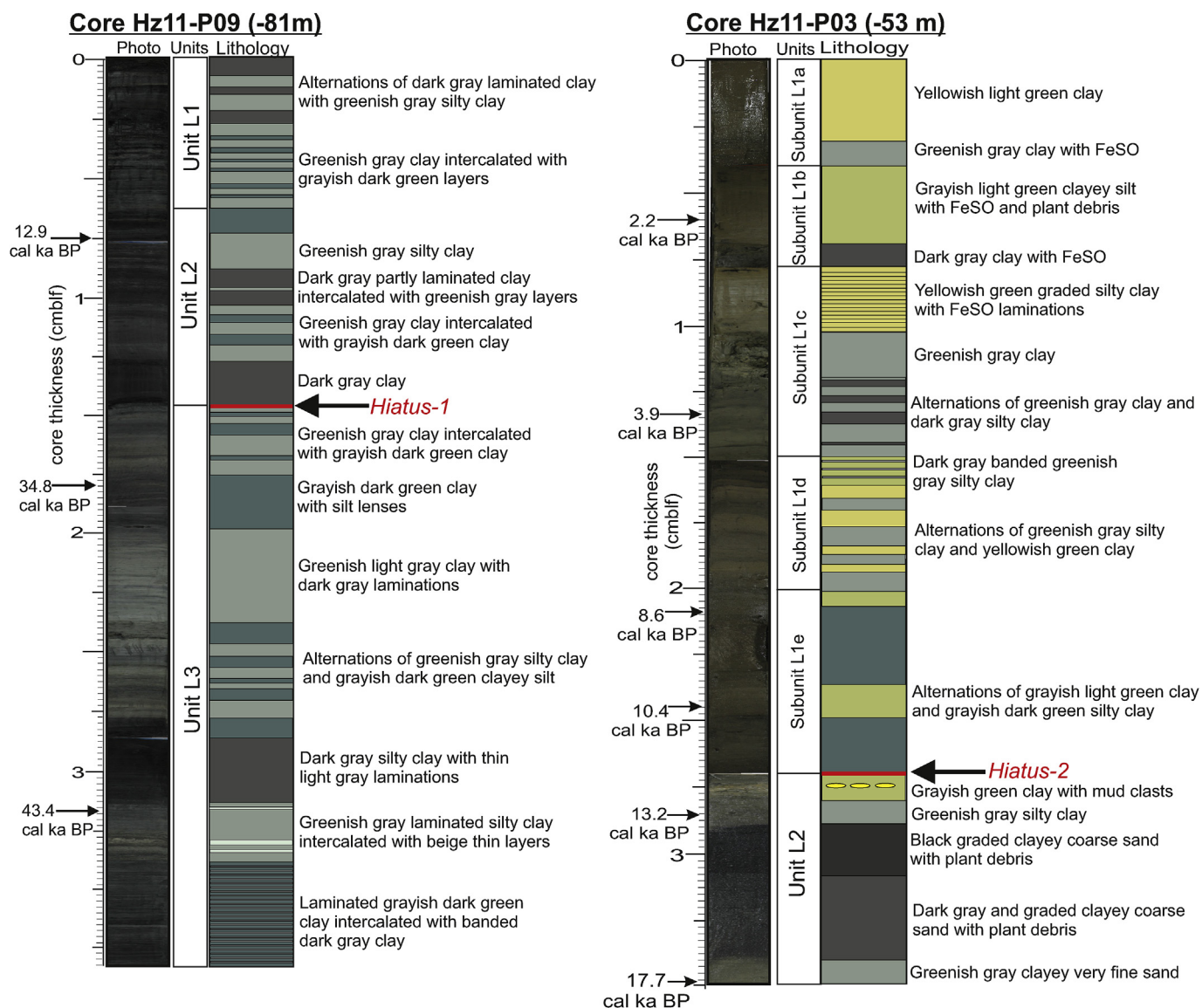


Fig. 4. Lithostratigraphic subdivisions of the studied cores Hz11-P09 and –P03 showing the late Pleistocene (units L3 and L2) to Holocene (Unit L1) sedimentary units that can be matched with the interpreted seismic profiles (in Fig. 3) (for locations of cores, see Fig. 2). Core Hz11-P03 contains two different units (L2 and L1) based on the lithologic and textural variations that are well correlated with equivalent units (units S2 and S1) in the seismic profile (see Fig. 3b). This shallower core was previously described by Eriş (2013), who subdivided the Holocene unit (L1) into five subunits, labelled as L1e to L1a. A remarkable lithologic change between units L2 and L1 represents a hiatus (Hiatus-2). In core Hz11-P09, the late Pleistocene units L3 and L2 is divided by a hiatus (Hiatus-1), corresponding to reflection surface SB-6 in the seismic profile (see Fig. 3a). The Holocene unit L1 in the core seems to be highly thinner comparative to the same unit in core Hz11-P03, implying a low resolution of sedimentary record. In the studied cores, ^{14}C datings are expressed as calibrated ages (without reservoir effect) that are also expressed in Tables 2 and 3

P09 was determined using X-ray diffraction (XRD) analysis with 15 cm resolution. Bulk mineral compositions of the samples from the core were determined using the procedure described by Gündoğdu (1982). The characteristic peak intensities of minerals were normalized to the reflection of dolomite.

3.3. Radiocarbon analysis

AMS (accelerator mass spectrometry) radiocarbon datings were carried out at the Woods Hole NOSAMS. The in-situ unbroken mollusc shells, some with closed valves, were used for radiocarbon analysis (Tables 2 and 3). The shells were carefully cleaned to remove any surface coatings, and then studied under microscope to ensure that they were diagenetically unaltered. Except the shells, a single ^{14}C age was obtained from wood piece sieved from the core

sediment. The ^{14}C ages are reported without reservoir correction, while we could not obtain sufficient wood materials from the same intervals with carbonate shell samples.

For core Hz11-P09, we used the Calib program (Stuiver et al., 2017) to calibrate the ages (Table 2), since we could not construct an age-depth model due to insufficient radiocarbon dates through the core. The Bacon software package was used to integrate the chronological information and develop an age-depth model for core Hz11-P03 (Ön et al., this issue). Instead of the “traditional” simple linear interpolation age-depth models, Bacon approach (Blaauw and Christen, 2011) utilizes bivariate monotone Markov process between predefined increments and produces “environmentally more inspired” models. Bacon calibrates the uncalibrated ages by using IntCal13 (Reimer et al., 2013) and constructs the age-depth model between the calibrated ages by Markov Chain Monte

Table 2

Radiocarbon ages obtained for core Hz11-P09. The ages are calibrated by using Calib program (Stuiver et al., 2017). mblf; meters below lake floor.

Site, water depth (m), dated material (lab.number)	Core locations	Core depth (mblf)	Radiocarbon ages (BP)	Calibrated median ages (cal BP)	Calib %95 confidence interval (cal BP)
Northern shelf, –81 m, ostracod (OS-110393)	Hz11-P09	0.75	11050 ± 45	12915	12786–13051
Northern shelf, –81 m, ostracod (OS-110394)	Hz11-P09	1.77	30900 ± 430	34835	34041–35685
Northern shelf, –81 m, ostracod (OS-110396)	Hz11-P09	3.18	39600 ± 1200	43437	41695–45389

Carlo iterations. In Table 3, we present the ages and their corresponding 95% confidence intervals, since they are used as ages for the dated depths.

4. Results

4.1. Seismic stratigraphy

In this study, seismic stratigraphic analyses of the high-resolution seismic reflection profiles in Lake Hazar are based on the previous preliminary interpretations of Eriş (2013). Accordingly, four seismic units (S4 to S1 from the bottom upward) (Fig. 3a and b) are bounded by the regional unconformities along the shelf and their correlative conformities (reflectors SB-8 to SB-5) in the lake basin. The current contribution in this paper exhibits an additional seismic data (Fig. 3a) tied to a radiocarbon-dated new sediment core (Hz11-P09 in Fig. 4), providing a much older chronology of sedimentary succession in Lake Hazar. The overall sedimentary succession in the seismic profiles begins at the base with Unit S4 that consists of progradational clinoforms characterized by acoustically strong internal reflectors (Fig. 3a and b). It is observed as prograding to a depth greater than –95 m. In the seismic profiles, Unit S4 is uncomfortably overlain by Unit S3 above the truncation surface SB-7, whereas it is conformably overlain by the same unit lakeward (Fig. 3a). On the lake margin, Unit S3 is marked by an aggradational architecture with a laterally continuous, regularly stratified package of highly-reflective internal configuration (Fig. 3a). On the deep basin as whole, it appears as a mud drape consisting of highly reflective comfortable parallel beds (Fig. 3a and b). Core-to-seismic correlation on the eastern part of the lake shows that core Hz11-P09 penetrates the most of Unit S3 (Figs. 3a and 4).

Unit S3 in the seismic profiles is separated from the overlying Unit S2 by reflection surface SB-6, which is a truncation surface on the lake margin with a maximum water depth of –95 m (Fig. 3a and b). Unit S2 over reflector surface SB-6 extends lakeward as shallow as –51 m. The seismic stratigraphic architecture of Unit S2 exhibits lenticular wedge of the truncated progradational clinoforms (Fig. 3a). Core-to-seismic correlations reveal that Unit S2 is recovered by two studied cores (Fig. 3a and b). In the seismic profiles, Unit S2 is overlain by an acoustically transparent youngest sedimentary unit (Unit S1), the two being separated by surface SB-5 (Fig. 3a and b). This surface is described as a shelf crossing unconformity that was previously assigned to the formation of –73 m

paleoshoreline in the lake (c.f., Eriş, 2013). In the seismic profiles, Unit S1 is the youngest depositional unit (Fig. 3a and b), which was previously subdivided into five seismic subunits (S1a–S1e) (Eriş, 2013).

4.2. Core lithologies

We mainly focused on two piston cores retrieved from Lake Hazar, having water depths of 53 m and 81 m, (Fig. 4; Table 1). Although core Hz11-P03 was previously described by Eriş (2013), one additional new piston core Hz11-P09 is also analyzed in the present study. According to the main lithological and textural variations in the cores (Fig. 4), three main depositional stages are differentiated in the lake, labelled from units L3 to L1 (from the oldest to youngest) based on Eriş's (2013) definition. Core-to-seismic comparison reveals that Unit L3 in the lower part of core Hz11-P09 correlates with a highly-reflective Unit S3 (Figs. 3a and 4). The general fining-upward nature of this unit exhibits variable lithologies with scarcity of molluscs shells, whereas abundant ostracods provides available radiocarbon dates through Unit L3 (Fig. 4; Table 2). The base of the unit mainly comprises finely laminated grayish dark green clay, alternating with banded dark green clay. In core Hz11-P09, the overlying greenish gray silty clay is intercalated with beige thin clay layers. AMS ^{14}C analysis of ostracod shell from the lower part of the unit gives an age of 43.4 cal ka BP (Fig. 4; Table 2). In Unit L3, a very dark gray homogenous silty clay layer is appeared between 313 mblf and 278 mblf. This darker layer is sharply overlain by alternations of greenish gray silty clay and grayish dark green clayey silt. Another sharp lithologic change in core Hz11-P09 appears above 238 mblf as greenish light gray clay comprising darker laminations. The uppermost part of Unit L3 in the core consists of grayish dark green clay alternated with a thinner greenish gray clay including a few silty lenses (Fig. 4). An AMS ^{14}C analysis of ostracod shell from the upper part of the unit gives an age of 34.8 cal ka BP (Table 2).

In the studied cores, Unit L2 is equivalent to Unit S2 in the seismic profiles (Figs. 3 and 4). The visual lithologic observation on core Hz11-P09 reveals an erosional boundary between units L3 and L2 at 146 mblf. According to matching core depth with sub-bottom depth in the seismic profile (Fig. 3a), this remarkable surface appears in the seismic profile (Fig. 3a) as truncational surface (SB-6) that could be an evidence of a hiatus in the sedimentary record (*Hiatus-1* in Fig. 4). Unit L2 in core Hz11-P09 displays variable

Table 3

Radiocarbon ages for core Hz11-P03. The ages in italic represent the additional ^{14}C dates obtained by this study, whereas the others were previously expressed by Eriş (2013). The ages are calibrated by IntCal13 (Reimer et al., 2013) in the Bacon software package. mblf; meters below lake floor.

Site, water depth (m), dated material (lab.number)	Core locations	Core depth (mblf)	Radiocarbon ages (BP)	Bacon weighted mean ages (cal BP)	Bacon %95 confidence interval (cal BP)
Northern shelf, –53 m, mollusc (OS-106010)	Hz11-P03	0.59	2240 ± 45	2206	2016–2342
Northern shelf, –53 m, wood (OS-97568)	Hz11-P03	1.35	3560 ± 45	3936	3722–4295
Northern shelf, –53 m, mollusc (OS-106006)	Hz11-P03	2.08	7800 ± 70	8586	8073–9051
Northern shelf, –53 m, mollusc (OS-106002)	Hz11-P03	2.44	9240 ± 70	10415	10190–10684
Northern shelf, –53 m, mollusc (OS-96328)	Hz11-P03	2.85	11400 ± 50	13211	12926–13437
Northern shelf, –53 m, mollusc (OS-96329)	Hz11-P03	3.48	14600 ± 80	17690	17214–18030

lithologies, ranging from partly laminated greenish gray silty clay to grayish dark green clay with intervening of dark gray clay layers (Fig. 4). In the upper part of the unit, AMS ^{14}C analysis of ostracod shells gives an age of 12.9 cal ka BP (Fig. 4; Table 2). The same unit with different lithologies was previously observed by Eriş (2013) in core Hz11-P03 (Fig. 4). Radiocarbon ages of 17.7 cal ka BP and 13.2 cal ka BP were previously obtained from Unit L2 (c.f. Eriş, 2013, Fig. 4; Table 3).

In core Hz11-P03, units L2/L1 boundary at 269.5 cmblf (Fig. 4) correlates with the major unconformity surface SB-5 in the seismic profile (Fig. 3b), thus it may represent another hiatus (*Hiatus-2*) in the cores. The lithologic and textural variations of Unit L1 in core Hz11-P03 were previously described by Eriş (2013), who subdivided five lithologic subunits (subunits L1e to L1a from bottom to top) based on the core-to-seismic correlation. In comparison to this subdivision, the core depths of the Holocene subunits are revised in the present study by carefully matching with sub-bottom depth in seismic profiles, thus, it provides a more accurate thickness assignment of the subunits in the core (Fig. 4). In the present study, additional AMS ^{14}C analyses (10.4 cal ka BP, 8.6 cal ka BP and 2.2 cal ka BP; Table 3) from Unit L1 in core Hz11-P03 provide precise timing of the Holocene subunits.

4.3. Chronostratigraphy of depositional units

In the present study, determination of the depositional units with an accurate chronology in the studied cores is necessary for precise core-to-seismic correlation and dating the key reflector surfaces of the main sedimentary units in the seismic profiles. According to cores chronologies in the present study, units L3 and L2 were deposited during the late Pleistocene, whereas Unit L1 represents the Holocene. In addition to Eriş (2013), three extra radiocarbon dates from core Hz11-P03 (Table 3) allow to construct a robust age-depth model for the last ca. 18 cal ka BP (Fig. 5) (Ön et al., 2017; this issue). Accordingly, approximate sedimentation rates for the pre-Holocene (between 17.7 cal ka BP and 11.7 cal ka BP) and Holocene period (after 11.7 cal ka BP) could be estimated to be $\sim 0.13 \text{ mm/yr}^{-1}$ and 0.25 mm/yr^{-1} , respectively (Fig. 5). The calculated sedimentation rate in the core is the lowest (0.15 mm/yr^{-1}) during the middle Holocene (8.1–4.2 cal ka BP), and the highest (0.3 mm/yr^{-1}) during the late Holocene (last 4.2 cal ka BP). In comparative to core Hz11-P03, the older chronology of the Hazar Lake sediments is obtained by three radiocarbon dates from units L3 and L2 in core Hz11-P09 (Fig. 4; Table 2). Based on the radiocarbon ages from Unit L3 (Fig. 4), the sedimentation rate is estimated to be $\sim 0.15 \text{ mm/yr}^{-1}$. Therefore, the base of Unit L3 in core Hz11-P09 could be assigned to an age of ca. 48 cal ka BP that implies deposition of this unit during Marine Isotope Stage 3 (MIS-3) (Figs. 4 and 6). However, the stratigraphic position of Unit S3 in relation with the younger units (S2 and S1) in the seismic profiles supports our age assignment (Fig. 3a and b). The calculated sedimentation rate ($\sim 0.15 \text{ mm/yr}^{-1}$) for the pre-Holocene suggests that the timing of the oldest hiatus (*Hiatus-1*) between units L3 and L2 in core Hz11-P09 is younger than 34.3 cal ka BP, coinciding with the early MIS-2 (ca. 29–23 cal ka BP). On the basis of correlative stratigraphy in the seismic profiles together with AMS ^{14}C datings in the cores, Unit L2 was deposited during the later part of MIS-2 (23–11.7 cal ka BP; Fig. 6), including the Last Glacial Maximum (LGM), Heinrich event 1 (H1), Bølling/Allerød (B-A) and Younger Dryas (YD) climate periods. Except the LGM, the timings of other climate events in Lake Hazar could be estimated by the age-depth model (Figs. 5 and 6) (Ön et al., this issue). Thus, the H1 in Lake Hazar spans during 17.7–14.8 cal ka BP, whereas the B-A period took place during 14.8–12.9 cal ka BP (Fig. 5). The YD, the last climate period prior to the Holocene, prevailed in Lake Hazar during 12.9–11.7 cal ka BP. When we

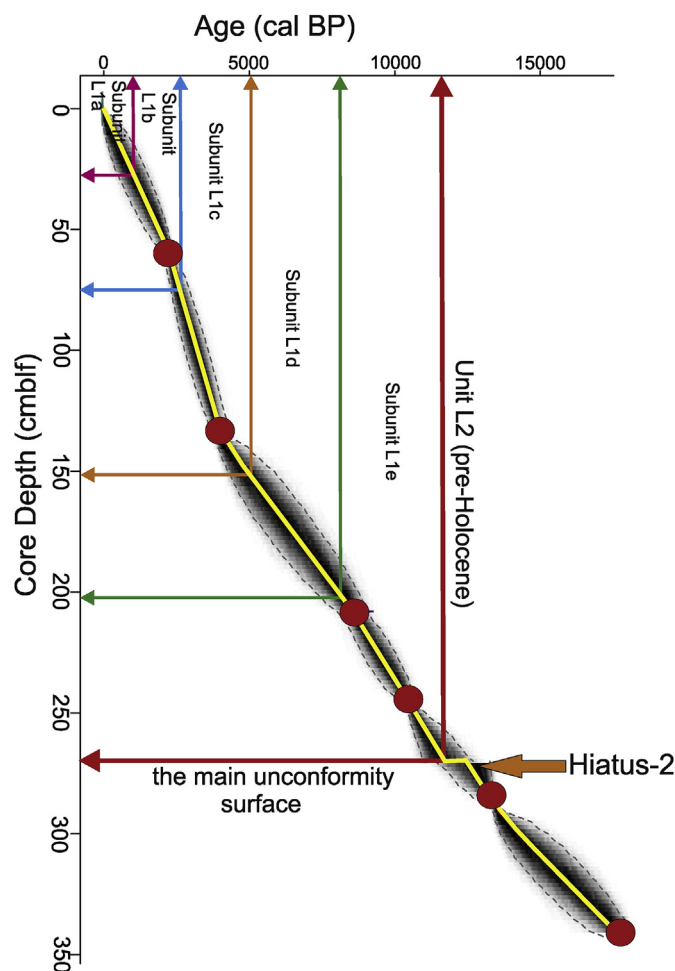


Fig. 5. An age-depth model of core Hz11-P03 is reconstructed on the basis of Bacon approaches (Ön et al., this issue), providing a promising chronostratigraphic subdivisions of the late Pleistocene to Holocene sedimentary units (L2 and L1) in Lake Hazar. The age-depth model dates precisely the Holocene subunits (see Fig. 6), and the late Pleistocene-Holocene boundary that appears as a hiatus (*Hiatus-2*) during 12.4–11.7 cal ka BP. Red dots represents six AMS ^{14}C dates obtained by Eriş (2013) and this study (see Table 3). Bacon calibrates the uncalibrated ages by using IntCal13 (Reimer et al., 2013) and constructs the age-depth model between the calibrated ages by Markov Chain Monte Carlo iterations. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

compare the ages of climate events in core Hz11-P03 with ones obtained from the Anatolian lakes (calibrated ages with reservoir effects), a few inconsistencies appear with differences ranging between ca. 100–400 years, and this could be an approximate reservoir effect for Lake Hazar. The most possible reason for such a short reservoir effect in the lake may be an occupation of mainly plutonic and volcanic and volcano-clastic rocks in the catchment areas of Lake Hazar (e.g., Erdem, 1987), implying the restricted source rocks for supplying high amount of old carbon to the lake.

The late Pleistocene-Holocene transition (units L2/L1 boundary; Fig. 4) in Lake Hazar is precisely dated in core Hz11-P03 at 11.7 cal ka BP based on the age-depth model (Fig. 5). This lithologic boundary in the core is defined as hiatus (*Hiatus-2*), covering a 700 year (between 12.4 cal ka BP and 11.7 cal ka BP) of non-deposition or erosion prior to the deposition of the overlying Unit L1. In comparison to Eriş (2013), the more accurate age assignments of the Holocene subunits (L1e to L1a) in core Hz11-P03 could be provided by the age-depth model (Fig. 6). Thus, it allows us to correlate their ages with the timings of formal division of the Holocene (e.g., 'early', 'middle' and 'late'; c.f. Walker et al., 2012). Such

SEISMIC REFLECTIONS	Lithostratigraphic Units		Seismic Units	Reflection characteristics	FACIES	
					On lake margin	Basin
SB-1 (1.4 cal ka BP)	Unit L1	Subunit L1a	Unit S1		Mud drape with highly-reflective horizontal beds	Mud drape with highly-reflective horizontal beds
		Subunit L1b			Clinoforms with strongly-reflective beds	Mud drape with highly-reflective horizontal beds
		Subunit L1c			Clinoforms with strongly-reflective beds	Mud drape with highly-reflective horizontal beds
		Subunit L1d			Clinoforms with moderately-reflective beds	Mud drape with highly-reflective horizontal beds
		Subunit L1e			Mud drape with highly-reflective horizontal beds	Mud drape with highly-reflective horizontal beds
	Unit L2		Unit S2		Mud drape with highly-reflective progradational clinoforms	Mud drape with horizontal beds
	Unit L3		Unit S3		Mud drape with highly-reflective aggradational clinoforms	Mud drape with horizontal beds
	Uncovered		Unit S4		Mud drape with highly-reflective progradational clinoforms	Mud drape with horizontal wavy beds

Fig. 6. Detailed core-to-seismic correlations in the present study allow us to reconstruct the main chronostratigraphic units, covering the time period of the last 48 cal ka BP. Only units L3 to L1 are penetrated by cores. The given ages for units L2 and L1 are calibrated on the basis of our age-depth model of core H211-P03 (see Fig. 5). According to the overall core chronologies, the Holocene sedimentary Unit L1 corresponds to MIS-1, whereas the late Pleistocene units L2 and L3 represent MIS-3 and MIS-2 periods. On the basis of calculated sedimentation rates from the studied cores, seismic Unit S4 is estimated to older than 48 cal ka BP. The facies associations of the Holocene subunits (L1e to L1a) depends on the seismic data previously documented by Eriş (2013).

comparisons together with variations in the multi-proxy data in the core provide us to better evaluate paleoclimatic reconstruction during different periods of the Holocene.

4.4. Physical, geochemical and mineralogical properties of the sediment cores

According to general trends of multi-proxy profiles through the oldest Pleistocene unit (L3) in core H211-P09, the marked variations are documented in distinct layers (Fig. 7). The lowermost laminated layer in the core (383–337 cmlbf) is represented by higher values in MS, Fe and K, whereas Ca/Ti and Sr/Ca inherently indicate lower ratios. The overlying layer above 337 cmlbf is marked by a sharp increase in Ca/Ti with strong fluctuations, in contrast to Fe, K and MS values. Until 287 cmlbf in the core, a higher dolomite content positively correlates with a gradual increase in Sr/Ca ratio, whereas TOC content exhibits general higher values with strong fluctuations similar to Ca/Ti profile (Fig. 7). The upper layer of Unit L3 (287–175 cmlbf) with a coarser-grained texture is likely recognized by the highest values in Fe and K as well as MS (Fig. 7). In contrast, Ca/Ti and Sr/Ca exhibit lower values as well as dolomite and TOC contents. The rest of the unit above 175 cmlbf is chiefly marked by gradual increases in Sr/Ca and Ca/Ti, although an abrupt increase in Fe value is documented in somewhere within this layer (Fig. 7). In core H211-P09, the late Pleistocene units L3/L2 boundary (*Hiatus-1*) at 146 cmlbf is determined by abrupt shifts in the multi-proxy records, indicating an almost maximum Sr/Ca and dolomite content together with a higher Ca/Ti until 87 cmlbf (Fig. 7). In Unit L2, abrupt increases in TOC content and Ca/Ti with strong fluctuations are noticeable between 87 cmlbf and 61.5 cmlbf, whereas a remarkable decrease in Sr/Ca positively correlates with dolomite content. In this part of Unit L2, a slight increase in MS value

positively correlates with Fe and K values.

In contrary to Eriş (2013), Fe and K profiles are added for core H211-P03 to better correlation with the other paleo-proxies in the core (Fig. 8). The H1 event in the lower part of Unit L2 (352–308 cmlbf) is accompanied by gradual increases in MS, Fe and K values in contrast to Ca/Ti and TOC. In core H211-P03, the main interval of the B-A period until 289 cmlbf is represented by a maximum MS value that positively correlates with Fe and K. These overall trends are consistent with a coarsening-up nature of the Unit L2 sediments in the core (Figs. 4 and 8). The later part of the B-A (289–280 cmlbf) is marked by opposite trends in Fe, K and Ca/Ti profiles. The YD period in the core (280–269 cmlbf) is determined by a sharp increase in Ca/Ti, whereas Fe and K profiles exhibit lower values with strong fluctuations (Fig. 8). In Unit L2, TOC content from the B-A to YD exhibits a general increasing trend with slight fluctuations.

In core H211-P03, the lower part of the early Holocene (269–236 cmlbf) in subunit L1e is chiefly marked by a sharp decrease in Ca/Ti (Fig. 8). The most prominent changes upward in Unit L1 are noticeable above 236 cmlbf with sharp increases in Fe and K values, although vice versa appears upward in subunit L1e. Within the same interval (236–222 cmlbf), Ca/Ti reaches almost its lowest value, then begins to increase gradually until 202 cmlbf in the core (Fig. 8). The early Holocene as whole is represented by a variable TOC content with a general increasing trend until the middle Holocene. The main part of middle Holocene (202–151 cmlbf) in core H211-P03 is represented by considerably high Ca/Ti with strong fluctuations, indicating negative correlations with K and Fe values. Another remarkable change is observed in TOC content that reaches the maximum value through the whole core (Fig. 8).

The transition from the middle to late Holocene (151–122.5 cmlbf) in core H211-P03 indicates high Fe, K and MS values in subunit L1c comparative to the earlier period, whereas Ca/Ti is

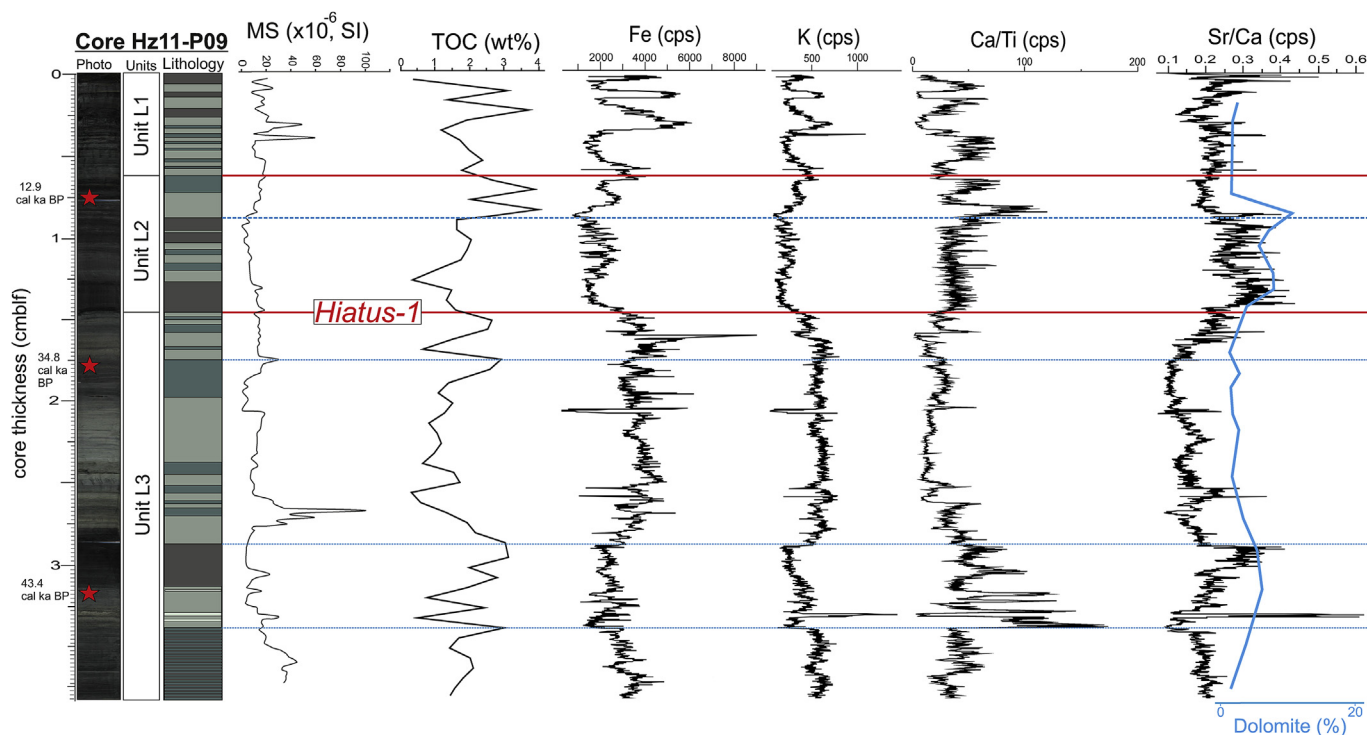


Fig. 7. Correlative plots of magnetic susceptibility (MS), Total Organic Carbon (TOC), Ca/Ti and Sr/Ca ratios and the other XRF analyses of Fe and K through core Hz11-P09. In the figure, dolomite content is added to compare with the other proxies. Red stars represents AMS ^{14}C ages (see Fig. 4 and Table 2) obtained from the core in this study. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

considerably low as well as TOC content (Fig. 8). The earlier period of the late Holocene (between 122.5 cmblf–106 cmblf) is marked by a sharp increase in Ca/Ti that almost reaches its maximum value within subunit L1c. In contrast, Fe and K values are considerably low as well as MS value. In the late Holocene, the highest Fe and K values are recorded between 106 cmblf and 57 cmblf (subunits L1c and L1b) in contrast to Ca/Ti (Fig. 8). In the same part of the late Holocene, the reverse trends in Fe, K and Ca/Ti are observed in two distinct layers (85–77.5 cmblf and 56.5–45.5 cmblf). The rest of the late Holocene until 8 cmblf in the core is represented by gradual increases in Fe and K in subunit L1a that negatively correlate with Ca/Ti and MS value (Fig. 8). During the late Holocene, the overall TOC content decreases until 56.5 cmblf, and then increase until the core top.

5. Discussion

5.1. Pre-Glacial period (>29 cal ka BP)

Analyses of seismic reflection profiles and cores together with multiproxy records provide detailed record of the lake level fluctuations and insight into paleoclimate of the Eastern Anatolia during the late Pleistocene to Holocene. The water level changes in Lake Hazar were likely controlled by the local climate governed by the regional evaporation–precipitation balance. Of course, the local tectonic processes related to the East Anatolian Fault Zone (EAFZ) can influence the interplay of lake level rise and sediment supply, but the quantification of tectonic control over the last ca. 48 cal ka BP is difficult to estimate by using paleo-proxy records. The seismic stratigraphic evidence from the oldest seismic unit (S4) indicates progradational clinoforms, implying decreasing lake level prior to 48 cal ka BP (early MIS-3) due to drier climate condition. However, the truncation of these clinoforms over the reflection surface (SB-7)

could be traced at the lake margin as deep as –95 m (Fig. 3a), showing the lowest water level in Lake Hazar. The later period of MIS-3 (ca.48–29 cal ka BP) was accompanied by a deposition of transgressive Unit S3 that is typically characterized by an aggradational architecture (Fig. 3a). The reflector surface (SB-6) truncating the top of Unit S3 extends as shallow as –55 m in the seismic profile (Fig. 3b), suggesting a considerable lake level increase due to a wetter climate. Such lake level rises during this period have been also documented from Lake Van (Kuzucuoglu et al., 2010; Çukur et al., 2014), Lake Zeribar and Lake Urmia (Synder et al., 2001; Djamali et al., 2008). On a global scale, general warm climate condition during this period of MIS-3 has been previously reported from paleoclimate records within the Mediterranean region (Martrat et al., 2004; Vogel et al., 2010).

Although general seismic configuration of Unit S3 exhibits transgressive lake level (Fig. 3a), high climate variability in Lake Hazar is documented by the multi-proxy records in core Hz11-P09 (Fig. 7). The general trends of elemental profiles allow us to distinguish four alternating wet and dry climate intervals. Similar climate variability with the Greenland interstadials and stadials (Dansgaard-Oeschger events; D-O and Heinrich events) during MIS-3 was also documented in the NGRIP data (NGRIP, 2004). The affects of the D-O events on the Anatolian and Near East climates have been also documented by the stable isotope records of Sofular Cave in western Black Sea coast of Turkey (Fleitmann et al., 2009; Göktürk et al., 2011) and Soreq cave in Israel (Bar Matthews et al., 2000). Considering a constant sedimentation rate ($\sim 0.15 \text{ mm/yr}^{-1}$) for Unit L3 in core Hz11-P09, the earlier wet interval (383–337 cmblf) took place prior to ca. 45 cal ka BP, whereas the later one (287–175 cmblf) prevailed during 41–34.3 cal ka BP (Fig. 7). These both wet intervals during MIS-3 are marked in the core by higher values in detrital-input proxies (MS, Fe and K), whereas lower Sr/Ca as well as dolomite content support the increased humidity. The

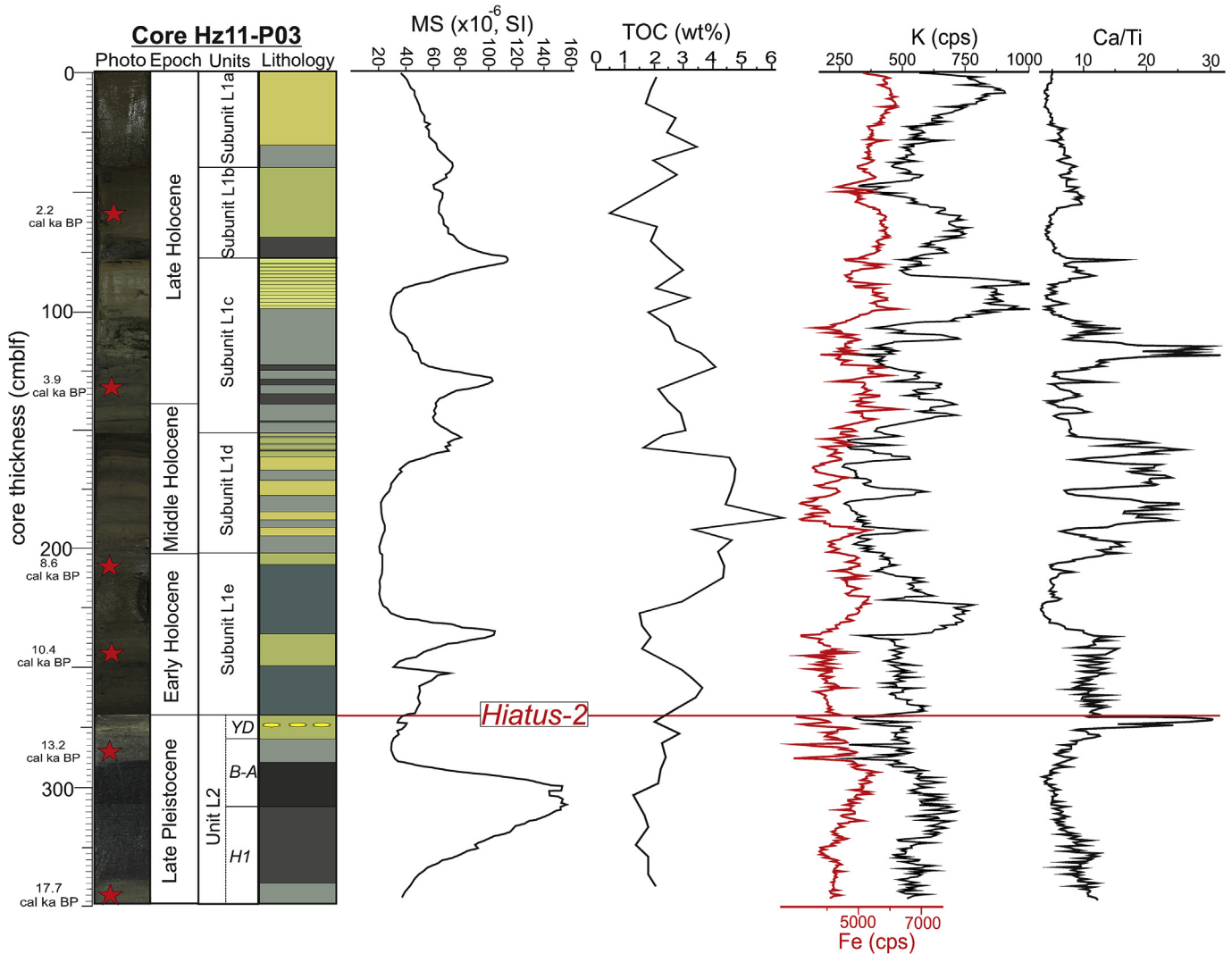


Fig. 8. Correlative plots of magnetic susceptibility (MS), Total Organic Carbon (TOC), Ca/Ti and the other XRF analyses of Fe and K through core Hz11-P03. The late Pleistocene Unit L2 in the core contains Heinrich event 1, Bolling-Allerod (B–A) and Younger Dryas (YD) climate periods. Red stars represents AMS ^{14}C ages (see Fig. 4 and Table 3) obtained from the core by Eriş (2013) and this study. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

lower TOC contents are notable during the most of wet intervals in MIS-3, although warmer and wetter climate can be expected to promote organic productivity in lakes (c.f. Çağatay et al., 2000; Tolun et al., 2002). The most possible reason for the insufficient organic matter preservation into the sediments could only be explained by highly fluctuating lake level, restricting formation of water stratification and anoxic deep-bottom water condition.

During the later period of MIS-3 (ca. 48–29 cal ka BP), the two shorter dry climate intervals with different intensities appear in the multi-proxy record of core Hz11-P09 (Fig. 7). In the lower interval (337–287 cmblf), the elevated dryness during ca. 45–41 cal ka BP is inferred particularly by gradual increase in Sr/Ca. In the same interval, relatively high dolomite abundances (Fig. 7) suggest high salinity and Mg/Ca conditions that can be linked to low lake level due to dry climate. The strongly-fluctuated Ca/Ti as well as TOC content within this dry interval are attributed to highly-fluctuating climate variability, presumably corresponding to the D–O events in MIS-3. The youngest dry interval (175–146 cmblf) in core Hz11-P09 between ca. 34.3 cal ka BP and 29 cal ka BP is similarly recognized by gradual increases in Sr/Ca and dolomite content that negatively correlates with the overall trends in Fe and K elements (Fig. 7).

At the onset of MIS-2 (ca. 29 cal ka BP), the evidence of considerable lake level decrease in Lake Hazar is documented by a truncational nature of the seismic surface SB-6 above Unit S3 (Fig. 3a). Since the maximum water depth of this surface (–90 m) is much below the depth (–82 m) of core Hz11-P09, the non-deposition and/or erosion at the core site could have produced a hiatus (Hiatus-1) between units L3 and L2 in the core (Fig. 4). Such time gap in the core stratigraphy may coincide with an abrupt climate shift due to the cold and dry climate in the early period of MIS-2 (29–23 cal ka BP). This conclusion can be supported by the elevated ^{18}O isotope value in NGRIP (NGRIP, 2004), implying maximum aridity in the northern hemisphere. The similar climate condition is also evidenced from the oldest and extensive glacial advances in the Anatolian mountains that have been dated prior to ca. 28 cal ka BP (Sarıkaya et al., 2009).

5.2. Glacial and late-glacial periods (29–11.7 cal ka BP)

The lowstand lake level during MIS-2 (29–23 cal ka BP) was followed later by deposition of Unit S2 in Lake Hazar with shallowest depth of –51 m in the seismic profile (Fig. 3b), implying a

comparatively higher lake level during the later period of MIS-2 (23–11.7 cal ka BP). The similar high lake levels were recorded in Lake Lisan (Kaufman et al., 1992; Bartov et al., 2002; Stein et al., 2010) and Konya Lake in central Anatolia (Roberts, 1983). Although deepening of the lake is inferred from the seismic data, Unit L2 in core Hz11-P09 partly exhibits considerable high maximum Sr/Ca ratio and dolomite content (Fig. 7), implying a dry climate during the LGM and H1. On the basis of the multi-proxy records in core Hz11-P03 (Fig. 8), the earlier period of H1 during 17.7–15.9 cal ka BP is represented by a low detrital input to Lake Hazar due to a drier climate, then it was followed by the increased humidity during 15.9–14.9 cal ka BP. In contrast, the H1 in Lake Van was documented by subaqueous prograding deltas at ca. –200 m as a consequence of considerable lake level drop (Damcı et al., 2012; Çukur et al., 2014). In core Hz11-P03, the maximum MS value together with Fe and K within Unit L2 in the core timely coincide with the earlier Bolling-Allerød (B-A) period (14.9–13.5 cal ka BP), displaying a clear evidence of the highest humidity in Lake Hazar. The black color of the lower unit L2 sediments with higher TOC content attests the increased lake level that would likely promote organic production and burial in the lake. The increased wetness during this period of B-A has also been documented from Sofular Cave at 14.6 ka cal BP based on increasing ^{13}C values (Fleitmann et al., 2009).

The return to a drier climate condition in Lake Hazar in the later part of the B-A (13.5–12.9 cal ka BP) is evidenced by gradual increases in Ca/Ti, reaching a maximum during the YD (Fig. 8). The existence of dry and cold climate during the YD in Lake Hazar was proposed by Eriş (2013) based on the μ -XRF Ca/Ti data in the same core (Fig. 8). This previous observation is supported by other paleoproxies (e.g., Fe and K) in this study. However, the progradational reflection configuration of Unit S2 in the new seismic data (Fig. 3a) can be attributed to a considerable lake level drop during the YD. According to Eriş (2013), the further lake level decrease during the YD was modulated by a stillstand, producing –73 m erosional terrace on the lake margin. Indeed, this conclusion strongly suggests an existence of subaerial erosion at the core site (–53 m), producing the hiatus (*Hiatus-2*) between units L2 and L1 in core Hz11-P03 (Fig. 4). The age-depth model of the core (Ön et al., this issue) placed this younger hiatus between 12.4 cal ka BP and 11.7 cal ka BP (Fig. 5). The similar cold and dry climate during the YD has been documented the other lake records from the central and eastern Anatolia (Landmann et al., 1996, 2011; Wick et al., 2003; Jones et al., 2007; Litt et al., 2009; Çağatay et al., 2014). The YD in the Eastern Mediterranean was expressed by the retreat of mesic forests and the spread of steppic vegetation, implying dry and cold climate conditions (Bottema, 1995; Rossignol-Strick, 1995). Similarly, rapidly decreasing ^{13}C values from Sofular Cave indicate the retreat of trees and shrubs at the southern Black Sea coast since 12.9 ka cal BP (Fleitmann et al., 2009).

5.3. Holocene (<11.7 cal ka BP)

In the Eastern Mediterranean as well as lakes and marshes elsewhere in the Anatolia, the cold and dry YD period was followed by a wetter early Holocene marked by increased precipitation (Bar-Matthews et al., 2000; Roberts et al., 2001; Jones et al., 2007; Çağatay et al., 2014; Dean et al., 2015; Miebach et al., 2016). The similar high precipitation at the onset of the Holocene in Lake Hazar is accompanied by a deposition of transgressive depositional unit (S1) (Eriş, 2013) at 11.7 cal ka BP (Fig. 3a and b). Recently, pollen record from Lake Hazar indicates that the expansion of deciduous *Quercus* at the beginning of early Holocene can be attributed to significant afforestation due to a high humidity (Biltekin et al., 2017). According to intensities of detrital-input proxies (MS, Fe

and K) in the lowermost Unit L1 (269–236.5 cmblf) (Fig. 8), the amount of precipitation during the earliest Holocene (11.7–10.1 cal ka BP) seems to be higher than the YD, but lower than the B-A period. However, the maximum precipitation in the early Holocene prevailed during 10.1–9.3 cal ka BP that is inferred by an abrupt positive excursion in Fe, K and MS values in the core (236–222 cmblf). On the other hand, the strongly fluctuated TOC contents can be attributed to changing organic productivity due to variations in lake level.

In Lake Hazar, the return to drier condition in the early Holocene took place during 9.3–8.2 cal ka BP based on the overall trends in Fe and K values in core Hz11-P03 (222–202 cmblf) (Fig. 8). In a global scale, the similar dryness at 9.3 ka BP in the Eastern Mediterranean was attributed to a decrease in annual insolation (Black et al., 2011). This remarkable drought was also reported in many other northern hemisphere regions as the 9.3 ka 'event' in China (Dykoski et al., 2005) and Oman (Fleitmann et al., 2004). In Lake Hazar, the main part of middle Holocene until 4.9 cal ka BP is represented by high climate variations with the wet intervals being shorter than the dry intervals based on general trend and intensities of Ca/Ti and detrital-input proxies (MS, Fe and K) in core Hz11-P03 (202–151 cmblf) (Fig. 8). Similarly, the multi-proxy records from the Mediterranean lakes indicate a series downward trending wet-to-dry oscillations during the middle Holocene (Roberts et al., 2011). However, a general aridification during the mid-Holocene was also inferred from $\delta^{18}\text{O}$ and $\delta^{13}\text{C}$ records of speleothems in Israel (Bar-Matthews et al., 2000), and by geochemistry and pollen data from crater-lake sediments in central Turkey (Roberts et al., 2001).

The crucial climatic shift from drier to wetter climate in Lake Hazar during the mid-to-late Holocene transition (4.9–3.7 cal ka BP) is evidenced by higher values in Fe, K and MS in core Hz11-P03 (151–122.5 cmblf) (Fig. 8). This period is also represented by higher primary productivity and efficient organic matter preservation comparative to the later period, although it seems still low in comparison to the middle Holocene. The most intense dry period appears between 3.7 cal ka BP and 3.3 cal ka BP that is marked by the maximum Ca/Ti throughout the core (122.5–106 cmblf) (Fig. 8). The same arid interval in Lake Hazar correlates with similar hydrological conditions in Lake Van, Lake Tecer and Lake Nar at 3.5 cal ka BP (Lemcke and Sturm, 1997; Kempe et al., 2002; Landmann and Kempe, 2005), and in the whole eastern Mediterranean during 3.65–3.6 cal ka BP (Roberts et al., 2011). On the basis of general trends in K and Fe values in core Hz11-P03 (Fig. 8), this short late Holocene drought in Lake Hazar was followed by the longer period of humidity between 3.3 cal ka BP and 2.1 cal ka BP (106–56.5 cmblf), whereas it was interrupted with a short intervening of dryness prevailed during 2.8–2.6 cal ka BP. The decreased TOC content in this interval (subunit L1c) indicates that the effective water stratification and efficient organic matter preservation were not supplied due to an insufficient lake level. During the latest Holocene, the another short but intense drought took place in the lake between 2.1 cal ka BP and 1.8 cal ka BP that can be deciphered by sharp negative excursions in Fe and K values in the core (55–47.5 cmblf) (Fig. 8). The same dry interval was also proposed for Lake Tecer (Kuzucuoglu et al., 2011) and Nar Lake (Jones et al., 2006). In Lake Hazar, the rest of the late Holocene until 3rd millennium (the top 8 cmblf) is associated with an elevated detrital-input proxies (e.g., Fe and K values), implying an increasing humidity.

6. Conclusions

This study, using seismic and multi-proxy data obtained from two piston cores, contains the first attempt to document the climate and lake level changes in Lake Hazar extending until marine isotope stage 3 (MIS-3). Adjustment of seismic units correlated

with the radiocarbon-dated chronostratigraphic units in the cores provides a basis for a chronological discussion of four main sedimentary units (units S4 to S1 from the oldest to the youngest), each subject to distinctive hydrological and sedimentological processes due to fluctuating lake level. The progradational facies of the oldest depositional unit indicates a considerable lake level drop to -95 m prior to 48 cal ka BP. The multi-proxy record from the deepest piston core suggests a high variable climate oscillations during the later period of MIS-3 (ca. 48–29 cal ka BP), when the lake level increased above -55 m. The onset of MIS-2 is accompanied by a maximum lake level lowstand at ca. -90 m prevailed during 29–23 cal ka BP, which is assigned to a remarkable hiatus in the sedimentary record. In the later period of MIS-2 (after 23 cal ka BP), the lake level was comparatively high, even though evaporation/precipitation is still higher than most of MIS-3, implying a general evaporative condition. According to multi-proxy records together with age-depth model of the core, the maximum humidity in Lake Hazar occurred during 14.9–13.5 cal ka BP. In consistent to the previous lake records from the eastern Anatolia, the Younger Dryas (YD) period in Lake Hazar was cold and dry that is chiefly marked by the μ -XRF Ca/Ti data. The lowstand lake level during the YD was followed by the Holocene transgression at 11.7 cal ka BP due to increased precipitation. During the early Holocene, the maximum precipitation in the lake is documented between 10.1 cal ka BP and 9.3 cal ka BP based on the detrital-input proxies in the core. The middle Holocene is accompanied by highly fluctuated climate conditions dominated by longer dry intervals. The general wet climate trend during the late Holocene gave rise to lake level rises that were interrupted by short decreases due to intense dry climates during 3.7–3.3 cal ka BP, 2.8–2.6 cal ka BP and 2.1–1.8 cal ka BP, respectively.

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